

FAI Event Post-Mortem Governance

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

A Full Aspect Integration (FAI) event produces a record — in the shared substrate and in each participating Self’s home substrate — that carries governance-relevant evidence: how the six FAI dimensions performed under the event’s conditions, what conflict classes arose and how they were handled, what evolution feed content was generated and what was absorbed, what the governance health indicators showed, and where the cross-organizational agreement proved inadequate. If that evidence is not systematically reviewed and converted into governance improvements, it dissipates. This note formalizes the governance practice that prevents that dissipation: the FAI event post-mortem. The post-mortem review operates across five review areas — configuration calibration, conflict pattern analysis, evolution feed quality, governance health, and cross-organizational agreement gaps — and produces a governance improvement record within each participating Self’s home substrate. That record is the input by which FAI event experience enters the directed selection mechanism for governance infrastructure evolution. The note states the five review areas with operational content, describes the governance improvement record and its authorization requirements, explains the post-mortem’s role as directed selection input, identifies the bilateral post-mortem advantage, establishes timing requirements, states the unreviewed accumulation anti-pattern, and provides an operational test.

1. D2.39 as operational decomposition of D1.27 and D2.35

D1.27 commits that accumulated FAI events across many CKS-governed Selves constitute a collective evolution mechanism — not merely a series of coordination episodes, but a

population-scale dynamic through which governance intelligence, operational patterns, and coordination schemas propagate and improve. The commitment is architectural: FAI events, when processed through each participating Self's home evolution machinery, produce changes to home governance infrastructure that carry into subsequent events and, at population scale, into the broader network of inter-Self relationships.

D2.35 formalizes the governance health indicator framework: the set of positive indicators and warning indicators that characterize governance quality within and across FAI events, enabling participating Selves' governance authorities to assess whether the governance infrastructure is functioning as intended.

D2.39 is the operational decomposition of both. It asks: given that D1.27 commits to a collective evolution mechanism and D2.35 provides the observational apparatus for governance health, what governance practice converts accumulated FAI event experience into actual governance improvement? The answer is the post-mortem review: a structured, five-area review of a completed FAI event that produces a governance improvement record as substrate content within each participating Self's home governance, and that feeds proposed improvements into the directed selection machinery for governance infrastructure evolution.

Without the post-mortem, D1.27's collective evolution commitment is aspirational rather than operational. Events accumulate; experience exists in the action-layer records produced by each event. But experience does not automatically become improvement. The post-mortem is the governed mechanism that performs the conversion.

2. The five review areas

A post-mortem review addresses five areas. Each area is distinct, covers different governance evidence, and produces different classes of improvement proposal. A post-mortem that omits any area is incomplete; the governance improvement record it produces has a corresponding gap.

2.1 Configuration calibration

The first review area examines how the six FAI dimensions — sharing scope, cardinality, persistence policy, cooperation/competition variant, provenance carry-over depth, and provenance preservation on internalization — were configured for the completed event, and whether that configuration was appropriate for what the event actually required.

The review asks: Did any dimension require amendment during the event (D2.38)? If so, which dimension was amended, what triggered the amendment, and what does the amendment reveal about the standing configuration's starting calibration? An amendment is evidence that the initial configuration did not match the event's conditions; the post-mortem review converts that

evidence into a proposed adjustment to the standing configuration (D2.21) for future events of the same class.

The review also asks in the negative: were there dimensions that were not amended but that, in retrospect, were set at suboptimal values — values that did not cause observable problems but that would have performed better at a different setting? This is the harder governance question; it requires the reviewing governance authority to reason about counterfactuals rather than only about observed failures. Both questions belong in the configuration calibration area.

2.2 Conflict pattern analysis

The second review area examines what conflict classes arose during the event and how they were handled across the three tiers: preservation within the shared substrate, resolution through orchestration rules, and escalation to humans.

The review asks: what conflict classes arose? For each class, which tier handled it? For escalation cases, what was the escalation outcome and what does it reveal about the completeness of the current orchestration rule set? For preserved conflicts, what was the downstream effect of carrying them through to evolution feed outputs and home substrate ingestion?

Conflict pattern analysis is the primary driver of new orchestration rule authoring (D2.15). When a conflict class arose that the current orchestration rules did not cover — forcing either preservation or escalation where orchestration resolution would have been more efficient — the review should produce a proposed new orchestration rule for that conflict class. When the same conflict class arose as in prior events with the same Selves, the review should ask whether the prior post-mortem's proposed rule was implemented and, if so, whether it performed as intended.

Over multiple events with the same Selves, conflict pattern analysis builds a cross-event picture of which conflict classes are recurring, which are event-specific, and which appear to be trend-driven — changing in character as the Selves' respective home substrates evolve. The accumulated picture is what makes the orchestration rule library for a given inter-Self relationship improve over time.

2.3 Evolution feed quality

The third review area examines what the event produced in terms of evolution feed outputs and what each participating Self's home governance did with those outputs.

The review asks: what evolution feed content was generated — what action-layer records, what DNA-layer content candidates? Of the DNA-layer content candidates, what did the receiving Self's governance authorize for absorption via directed selection, and what was declined? What

does the absorption/decline pattern suggest about whether the sharing scope configuration is well-matched to what the receiving Self actually finds governable and useful?

A high absorption rate against a broad sharing scope suggests the scope is well-calibrated. A low absorption rate — or a pattern of declining large classes of contributed content — suggests a mismatch: the contributing Self is sharing content the receiving Self's governance cannot or will not absorb, which is waste in the evolution feed. The post-mortem review converts the absorption/decline pattern into a proposed adjustment to sharing scope configuration.

The review also examines the action-feedback path: what action-layer records from the event are entering each Self's home action-feedback evolution machinery, and what proposals do those records appear likely to generate? This is necessarily prospective and uncertain at review time, but governance authorities benefit from flagging action-layer content that may generate proposals requiring immediate attention — particularly where the event surfaced edge cases or failure patterns that home orchestration rules do not yet cover.

2.4 Governance health indicators

The fourth review area examines the governance health profile of the completed event against the ten positive indicators and the warning indicators established in D2.35.

The review asks: which positive indicators applied? A complete application of all ten positive indicators is the benchmark for a well-governed event; partial application identifies which governance dimensions performed as intended and which did not. Which warning indicators fired? A warning indicator that fired once may be an event-specific anomaly; a warning indicator that fired in the same area as in prior events with the same Selves is a governance pattern that warrants structural attention.

The governance health profile serves two functions. First, it characterizes the governance quality of the specific event — useful for the participating Selves' governance authorities as an accountability record. Second, it feeds into the ongoing relationship assessment: is governance quality for this inter-Self relationship improving, stable, or degrading over multiple events? The trajectory matters as much as the point-in-time score. A deteriorating trajectory in governance health is itself a governance warning that the relationship's foundational configurations and agreements need attention before the next event.

2.5 Cross-organizational agreement gaps

The fifth review area examines whether the cross-organizational agreement governing the relationship (D2.34) proved adequate for everything the completed event required.

The review asks: were there escalation routing situations the agreement did not cover — situations where a conflict needed to surface to humans but the agreement did not specify

whose humans, through what channel, under what timeline? Were there audit rights situations where one Self needed access to information for accountability purposes but the agreement's audit rights provisions did not clearly authorize it? Were there dissolution or persistence policy questions the agreement left ambiguous?

Each identified gap is a proposed amendment to the cross-organizational agreement. Agreement gaps often appear only under operational stress — the agreement looks complete before events begin but reveals its silences when actual coordination requires something it did not anticipate. The post-mortem is how those silences become explicit, and how explicit gaps become amendment proposals rather than recurring friction.

3. The governance improvement record

The post-mortem review produces a governance improvement record — home governance content within each participating Self's home substrate. The record is authored under the reviewing Self's home governance authority; it is substrate content carrying all six Paper 1 commitments including provenance attribution, human governance, and the three rights of inspection, modification, and override.

The record contains four elements:

Review findings. A summary of findings for each of the five review areas, including the evidence base — references to the event's shared substrate record and to home-side action-layer records — and the governance authority's assessment of what each finding means for future events.

Proposed improvements. For each finding that warrants governance infrastructure change, an explicit improvement proposal: a proposed adjustment to the standing configuration for a specific dimension, a proposed new orchestration rule for a specific conflict class, a proposed amendment to a specific provision of the cross-organizational agreement. Proposals are specific enough to be accepted or declined as discrete governance decisions; they are not general intentions.

Governance authorization. For each proposed improvement, the governance authorization record: which authority has approved the improvement for implementation, what the implementation scope is (this Self only, or joint with the other participating Self), and what timeline applies. Proposed improvements that have not received governance authorization are flagged as pending rather than as approved.

Implementation tracking. For improvements authorized in prior post-mortem cycles that should now be implemented, the record documents implementation status — whether the prior improvement was implemented as approved and whether it produced the intended effect. Implementation tracking closes the feedback loop between post-mortem findings and

governance infrastructure change.

The governance improvement record is home governance content, not shared substrate content. Each Self produces its own record. The two records for the same event will overlap substantially — both Selves reviewed the same event — but they are authored independently under separate governance authority and may reach different conclusions about improvement priorities and authorization.

4. Post-mortem as directed selection input

The governance improvement record from a post-mortem is not merely an archive. It is an active input to the directed selection mechanism for governance infrastructure evolution.

Paper 2 establishes directed selection as one of three evolution mechanisms within a CKS-governed Self: governance-authorized DNA changes authored under human authority as a counterweight to mutation (LLM version changes) and action-feedback evolution (operational evidence feeding into DNA proposals). Directed selection is how governed, intentional change enters a Self's governance infrastructure.

Post-mortem proposed improvements are candidates for directed selection events applied to governance infrastructure. A proposed adjustment to a standing configuration dimension is a candidate for a directed selection event that modifies the standing configuration substrate. A proposed new orchestration rule is a candidate for a directed selection event that adds a rule to the orchestration rule set. A proposed cross-organizational agreement amendment is a candidate for a joint directed selection event operating under both Selves' governance authority.

The analogy to action-feedback evolution is instructive. Action-feedback evolution converts operational experience from the action layer into proposals for DNA change, subject to governance acceptance or rejection. Post-mortem review converts FAI event experience into proposals for governance infrastructure change, subject to the same governed acceptance or rejection. Both are experience-driven improvement mechanisms. Both are subject to human governance authority at the acceptance/rejection decision point. Both produce changes that are substrate content — durable, attributed, auditable.

The post-mortem is therefore the mechanism that connects individual FAI events to the collective evolution dynamic D1.27 anticipates. D1.27's collective evolution commitment requires that accumulated events improve the governance infrastructure through which future events operate. Post-mortem review — producing improvement records, proposing improvements, feeding proposals into directed selection, implementing authorized changes — is the operational chain that performs that improvement. Without the chain, accumulation is not evolution.

5. Timing and the bilateral advantage

Timing. Post-mortem reviews should occur after the event has dissolved and home evolution processing is underway — the evolution feed content has been identified, absorption/decline decisions are at least in initial stages — but before the next FAI event with the same Selves commences. This timing window is essential: if the next event begins before the prior post-mortem’s improvement proposals have received governance authorization and been implemented, the governance infrastructure entering the next event is the same as the one that served the prior event, regardless of what was learned. Improvements must be implemented between events to take effect. Governance authorities should treat the post-mortem completion, authorization, and implementation cycle as a precondition for scheduling the next event with the same Selves when the relationship is ongoing and improvements have been identified.

Bilateral advantage. The post-mortem review is most valuable when both (all) participating Selves conduct reviews and share findings. The bilateral advantage operates through the inspect right (D2.04): each Self holds the right to inspect the shared substrate record from the completed event, which means each Self can review the complete event record rather than only its own side. But inspect rights over the record do not automatically produce shared post-mortem findings. The sharing of findings requires either explicit exchange through the cross-organizational agreement’s information-sharing provisions, or governance authorization of sharing under the inspect right.

When both Selves share their post-mortem findings, the combined governance intelligence is qualitatively richer than either unilateral review. Self A observed the event from the perspective of its contributions and absorptions; Self B observed the same event from a different vantage. Conflict patterns A flagged may not be visible to B from B’s side of the record, and vice versa. Evolution feed quality assessments from A’s home governance — what B’s content looked like as potential absorption material — give B information B cannot generate from its own internal review. The bilateral picture of governance health, assembled from both Selves’ readings of the same ten indicators against the same event, is more complete than either reading alone.

Bilateral post-mortem exchange should be treated as a default practice when the cross-organizational agreement’s information-sharing provisions permit it. Where the agreement does not yet permit it, the absence of bilateral exchange provisions is itself a finding for the agreement gaps review area (§2.5) and a candidate amendment.

6. Anti-pattern: unreviewed accumulation

The failure mode to be named explicitly is unreviewed accumulation: FAI events accumulate between the same Selves over months or years, but no post-mortem reviews are conducted. The

governance infrastructure — standing configurations, orchestration rule sets, cross-organizational agreements — is never updated from the accumulated experience. Conflict patterns that first appeared in early events recur in later events because no new orchestration rules were authored. Standing configurations carry their initial calibration forward unchanged because no post-mortem ever produced a proposed adjustment. Escalations repeat the same routing ambiguities because the agreement was never amended.

Unreviewed accumulation is coordination without learning. Individual events may function acceptably — the shared substrate operates, evolution feed content is generated, Selves absorb and evolve — but the governance infrastructure does not improve from those events. The relationship between the two Selves does not mature. At population scale, the collective evolution dynamic D1.27 anticipates depends on individual Selves improving their governance infrastructure through experience. Unreviewed accumulation at scale is the population-level failure: many CKS-governed Selves accumulating many FAI events without any of those events feeding into governance infrastructure improvement. The coordination occurs; the evolution does not.

The post-mortem is not an optional enhancement for sophisticated deployments. It is the mechanism through which FAI governance infrastructure learns. Deployments that coordinate without reviewing are missing the learning half of the architecture.

7. Operational test

A deployment instantiates the FAI event post-mortem governance commitment if and only if the following are true:

For any FAI event completed more than one review cycle ago between two participating Selves, an observer with access to each Self's home governance substrate can locate a governance improvement record for that event that covers all five review areas: configuration calibration findings, conflict pattern analysis findings, evolution feed quality findings, governance health profile, and cross-organizational agreement gap analysis.

For any improvement proposal contained in that governance improvement record that received governance authorization, the observer can locate in the home substrate evidence of implementation: an updated standing configuration entry, a new or revised orchestration rule, an amended cross-organizational agreement provision, or a documented governance decision to decline implementation with rationale.

For any improvement proposal that did not receive governance authorization, the observer can locate either a pending-authorization flag or a declined record with rationale — confirming that the proposal entered governance review and received a disposition rather than simply expiring

without action.

A deployment that passes this test has operationalized the post-mortem governance practice. A deployment where the observer cannot locate governance improvement records for completed events — or can locate records that do not cover all five areas — has not operationalized the practice, regardless of whether post-mortem reviews were informally discussed. Governance commitment in the CKS sense is substrate content under human authority; discussion without a record is not governance.